

Quantum Technologies Research Project

January Report

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Project Goals & Metrics

Research Gap: No unified comparison of BB84 vs MDI-QKD across key metrics

**Key Rate vs
Distance**
(varying networks)

**Implementation
Cost**
(economic & energetic)

Scalability
(user count)

- Analyse both protocols across varying network topologies
- Identify parameter regimes & topologies where each excels
- Inform practical QKD deployment decisions

Upcoming milestones (from Project Outline):

1. **Mathematical Modelling**

- Note: re-ordering of expected milestone delivery timings

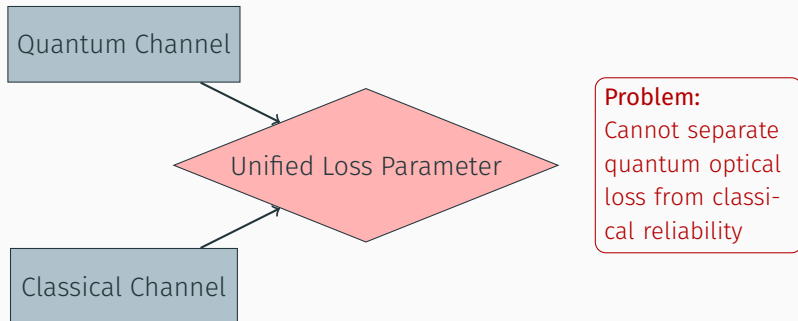
2. **Simulations Constructed**

3. **Prototype Simulation Results**

4. **Literature Review**

- Built basic QuNetSim BB84 simulator
- Outputs key rates with distance control
- Error reconciliation included
- Physical modelling begun
- No error correction/privacy amplification yet

Critical Issue: Unified classical/quantum loss model

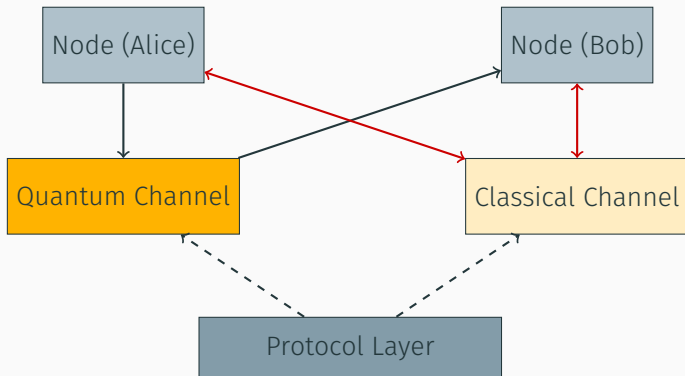


- Physical realism compromised for fair BB84/MDI comparison
- Motivated migration to NetSquid

- Adapted code from public repo of NetSquid protocol collection
- Outputs key rates with distance control
- Error reconciliation included
- Distance-dependent loss modelling by design
- No error correction/privacy amplification yet

Key advantage: Disaggregated quantum/classical channels

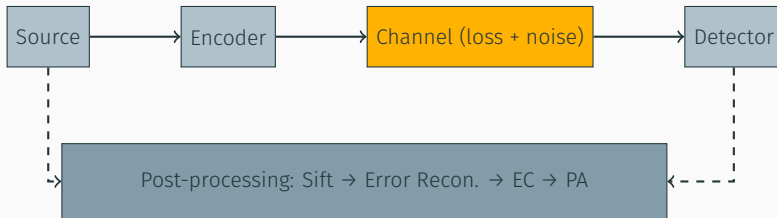
NetSquid - Architecture



- Separate, detailed channel models (loss, noise, timing)
- Event-driven simulation engine
- Fine-grained control over physical parameters

*Note extra node in MDI.

BB84 Simulator Design in NetSquid



Modular components:

- Source: photon generation
- Encoder: basis choice + polarisation
- Channel: distance-dependent loss, detector noise (controllable)
- Post-processing: sifting, error reconciliation, EC, PA

Critical Design Questions

Physical Modelling:

- What level of detector detail is needed for valid comparison?
- Full detector model vs simplified error channel?
- Adaptation of BB84 to fit MDI?

Validation & Metrics:

- Which baseline results validate the simulator?
- How to define "quantum cost" metric in practice?
- Parameter alignment: what must match between BB84 & MDI?

Scope & Priorities:

- Which topologies are most informative?
- Trade-off: physical realism vs computational tractability?
- What to drop if time-constrained?

Expected Key Rate Behaviour

Anticipated results:

- **BB84**: higher rate, drops faster
- **MDI**: lower rate, extends further

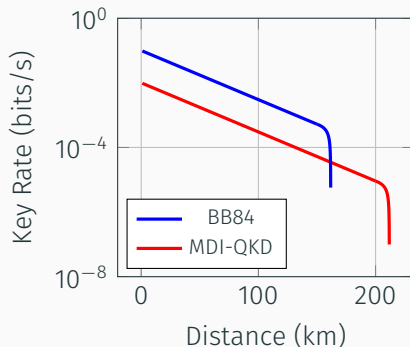
Topologies:

- Star minimises cost for BB84, full mesh maximises key rate
- Star doubly optimal for MDI

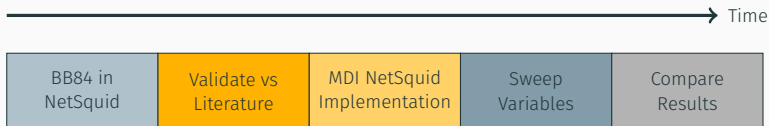
Scaling:

- Cheaper to add node to MDI
- Greater setup cost for MDI
- Run cost variation?
- Form of $\frac{d(\text{key rate})}{dn}$?

Goal: Quantify trade-offs



Roadmap: BB84 → Full Comparison



Implementation stages:

1. Finalise BB84 implementation in NetSquid
2. Validate against known rate-distance curves from literature
3. Adjust BB84 → MDI-QKD with symmetric setup (+ validate)
4. **Sweep distances, topologies and scale**
5. **Comparative analysis** across metrics

Completed

- QuNetSim BB84 prototype
- Identified QuNetSim limitations
- NetSquid architecture planned
- Initial NetSquid simulator tested

In Progress

- NetSquid BB84 refinement
- Parameter selection
- Validation benchmark identification
- Modelling assumptions documentation
- Planning for scaling modelling

Next immediate steps: Finalise NetSquid two-point BB84 and reproduce literature benchmarks; adapt to two-point MDI

Strategic decisions:

- Is the QuNetSim → NetSquid migration justified given timeline constraints?
- What constitutes sufficient validation for simulators and modelling?

Implementation priorities:

- Priority order: BB84 baseline → MDI baseline → topology variation → cost/scaling?
- Process for early-stage reviewing modelling assumptions?

Questions & Discussion

Design decisions and next steps